

Recurrence of afebrile status epilepticus in a population-based study in Rochester, Minnesota

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ABSTRACT Objective: To determine the risk of recurrence of status epilepticus (SE) in a population-based sample and to identify risk factors for recurrence. **Methods:** We ascertained all first episodes of afebrile SE in residents of Rochester, MN, through the Rochester Epidemiology Project's records-linkage system between January 1, 1965, and December 31, 1984. Information was collected on age, gender, duration, seizure type, etiology, therapeutic response to initial antiepileptic drug medication, and subsequent episodes of SE. **Results:** Among the 183 episodes of first afebrile SE, the risk of recurrent SE was 31.7% over a 10-year follow-up period. The risk of recurrence was about 25% for those with acute symptomatic SE, remote symptomatic SE, and idiopathic cryptogenic SE. Recurrence was 100% for those with progressive symptomatic SE. Female gender (rate ratio [RR] = 2.3, 95% CI = 1.1 to 5.0) and progressive symptomatic etiology (RR = 2.4, 95% CI = 0.6 to 8.9) increased the risk for recurrent SE. Both partial SE (RR = 0.5, 95% CI = 0.2 to 1.1) and good therapeutic response to the initial antiepileptic drug therapy (RR = 0.3, 95% CI = 0.1 to 0.7) were associated with a decreased risk of recurrent SE. **Conclusions:** Status epilepticus (SE) recurs in about one-third of individuals with a first episode of SE. Except for SE occurring in the setting a progressive brain disorder, the risk of recurrence is about 25%, regardless of the underlying etiology. Female gender and lack of response to the first antiepileptic drug medication after the initial episode of SE identify those individuals at greatest risk for recurrence. **NEUROLOGY 2007;69:73-78**

Status epilepticus (SE), a medical emergency, is usually defined as a seizure lasting at least 30 minutes.¹ It can occur in association with epilepsy and in the context of acute CNS or systemic insults.³ The incidence of SE is approximately 18 per 100,000 population among Caucasians³⁻⁷ and is higher in African Americans.³ Short-term mortality is high when SE occurs because of an acute insult.^{2,3} Among survivors, long-term mortality is greater than that expected in the population.⁸ Little is known about the frequency and risk factors for recurrent afebrile SE except in children.⁹⁻¹² Such information is important because it allows prediction of who is at higher risk for recurrence of this serious condition.

We report a population-based study of the recurrence of afebrile SE in Rochester, MN. We examine several potential predictors of recurrence, including age and gender of the participant, as well as factors associated with the initial SE, including etiology, seizure type, duration, and response to therapeutic intervention.

Supplemental data at
www.neurology.org

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METHODS Study population. Subjects with incident afebrile SE were ascertained through the records-linkage system of the Rochester Epidemiology Project. This system includes all contacts of Olmsted County residents with the medical system in southeastern Minnesota, including inpatient, outpatient, and home visits. Study subjects were all residents of Rochester, MN, identified because febrile seizure, acute symptomatic seizure, unprovoked seizure, or epilepsy was identified between 1955 and 1984.¹³ Charts of cases with any type of convulsive disorder were reviewed by two neurologists to identify those with SE and to classify SE by seizure type, etiology, and duration. Although ascertainment of convulsive SE is likely complete, there may be some underascertainment of nonconvulsive SE, using this methodology.

We defined SE as a single clinical seizure lasting more than 30 minutes or repeated seizures over a period of more than 30 minutes without intervening recovery of consciousness.¹ We excluded seizures lasting less than 30 minutes whether or not they were stopped by valium or other antiepileptic drug medication, multiple seizures in a 24-hour period each less than 30 minutes duration with intervening intervals of lucidity, and electroencephalographic (EEG) changes consistent with SE in the absence of clinical features.

Predictors of recurrent SE considered. For each resident with a first episode of afebrile SE between 1965 and 1984, we examined the risk of recurrent SE until February 27, 1996. Recurrent SE was ascertained through the medical records-linkage system. We examined age, gender, duration of initial SE, seizure type, etiology of initial SE, and therapeutic response to antiepileptic drug medication as risk factors for recurrence. Predictors were defined as follows:

Seizure type. SE seizure type was categorized by one of us (W.A.H.) based on criteria of the International League Against Epilepsy¹⁴ as generalized from onset, secondarily generalized, partial without generalization, myoclonic, or absence. Generalized convulsive SE included cases with generalized tonic, clonic, and tonic-clonic SE. Secondarily generalized SE included cases with generalized tonic, clonic, and tonic-clonic SE accompanied by focal features. Partial SE included cases with simple or complex partial seizures that did not generalize. Myoclonic SE included cases with bilateral or multifocal myoclonic jerks recurring at different intervals in time and in clusters; these are probably equivalent to “subtle SE” described by others.¹⁵ The classification of seizure type was based upon the description written by the attending medical staff, neurologists, nurses, and trained medical personnel such as EEG technicians and ambulance attendants. Generalized convulsive SE served as the referent group for multivariate analyses.

Etiology. Each episode of SE was reviewed to determine whether it was acute symptomatic or unprovoked, according to the criteria of Hauser et al.¹³ as follows:

Acute symptomatic. SE was considered acute symptomatic when it occurred in association with (within a week) of the onset of the following conditions: brain trauma, CNS infection, cerebrovascular disease, acute diffuse encephalopathy (primarily anoxia), and toxic/metabolic insults including alcohol or drug withdrawal. We classified febrile SE as a special class of acute symptomatic SE.

Unprovoked. SE was considered unprovoked in the absence of an identified acute insult and was further categorized as progressive symptomatic, remote symptomatic, or idiopathic/cryptogenic.¹⁶

Progressive symptomatic. SE was considered progressive symptomatic in the presence of nonstatic CNS conditions such as CNS tumors and degenerative neurologic diseases.¹⁶

Remote symptomatic. SE was considered remote symptomatic in the presence of a history of a CNS insult such as stroke, head trauma, or meningitis thought to lead to a static lesion and associated with an increased risk of epilepsy. The time between SE and the neurologic insult had to be more than 1 week but was usually months or years.

Idiopathic/cryptogenic. SE was considered idiopathic/cryptogenic in the absence of an acute precipitating factor or a history of a prior neurologic insult. This group served as the referent in the multivariate analyses.

Duration. Information on duration of SE was obtained by reviewing the ambulance, emergency room, and inpatient records (including physician notes, nursing records, and EEG information). Duration was categorized as 30 minutes to 2 hours, 2 to 24 hours, and 24 hours or longer. Duration of 30 minutes to 2 hours was used as the referent in the multivariate analyses.

Age. Age was categorized as <1 year, 1 to 20 years, 20 to 65 years, and 65 years or more. Age 1 to 20 years was used as the referent in the analyses.

Therapeutic response to anticonvulsants. Response to the first anticonvulsant given to control SE was categorized according to whether there was any suggestion that seizures stopped, even briefly. Therefore, patients were considered responders if SE stopped without any further seizure activity, and if SE stopped and then recurred, if SE stopped and further shorter seizures occurred. All other patients were considered nonresponders.

Gender. The risk of recurrent SE in females was compared with that in males.

Decade. The risk of recurrent SE was compared among individuals with a first episode of SE in 1965 through 1974 with those with a first episode in 1975 through 1984.

Statistics. We used Student *t* test to compare continuous variables and the χ^2 statistic to compare categorical variables.

Kaplan–Meier survival analyses¹⁷ were used to calculate the cumulative incidence of subsequent SE for all residents with an incident episode of SE. This analysis was repeated by etiology of the seizure, seizure type, duration of SE, age at first SE, and therapeutic response to the first antiepileptic drug medication for SE. Because the mortality after SE is high in the first 30 days, we also examined the risk of recurrent SE in the first 30 days and over the next 10 years among 30-day survivors. The log rank statistic was used to compare the risk of subsequent SE for subjects with and without the predictor. Subjects were followed until the date of last medical contact in southeastern Minnesota, the date of last southeastern Minnesota residency, or date of death.

Cox proportional hazards regression¹⁷ was used to examine the effect of predictors on the risk of subsequent SE. Predictors that were significant in the univariate model were retained in the multivariate model.

RESULTS There were 183 first episodes of first afebrile SE among Rochester residents between January 1, 1965, and December 31, 1984. SE was classified as acute symptomatic in 101 cases (55.2%) and unprovoked in 82 cases (44.8%) (table E-1 on the *Neurology* Web site; www.neurology.org). Acute symptomatic cases were due primarily to anoxic encephalopathy (*n* = 20, 19.8%), CNS infection (*n* = 17, 16.8%), and cerebrovascular disease

($n = 17$, 16.8%). Among unprovoked cases, 28 (34.2%) were idiopathic/cryptogenic, 17 (20.7%) were progressive symptomatic, and 37 (45.1%) were remote symptomatic. The most common antecedents of SE among remote and progressive symptomatic cases combined were degenerative brain disease ($n = 11$, 20.3%) and cerebrovascular disease ($n = 18$, 33.3%).

SE was generalized without focal features among 28.4% of cases ($n = 52$), secondarily generalized among 17.5% of cases ($n = 32$), and partial without generalization among 40.4% of cases ($n = 74$). Absence and myoclonic SE were less common than other seizure types, accounting for 3.3% ($n = 6$) and 10.4% ($n = 19$) of all first episodes of afebrile SE (table E-1).

A first recurrence of SE was found in 27 individuals by 2 years and in 33 individuals by 10 years. There were no recurrences after 10 years. There was a trend for the proportion with recurrence to increase with each additional recurrence: Eight individuals (24.2% of those with a second episode of SE) experienced a third episode of SE, four individuals (50.0%) experienced a fourth episode of SE, and two individuals (50.0%) experienced a fifth episode of SE.

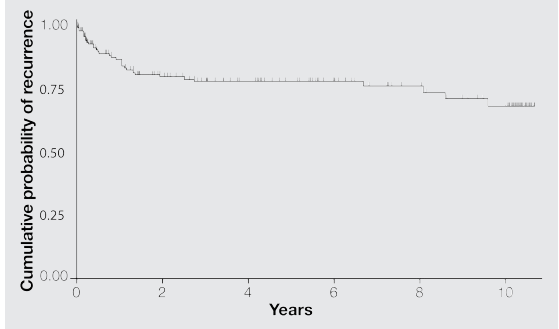
Overall risk of recurrence. The risk of recurrence of afebrile SE was 31.7% ($n = 33$) within 10 years of the incident episode of SE (figure). The risk of recurrence was 2.4% during the first 30 days after the first episode of SE with no difference by etiology. Most recurrences (81.8%) occurred during the first 2 years after the incident episode of SE. The risk of recurrent SE was 13.1% at 1 year and 20.3% at 2 years. Among 30-day survivors, the risk for recurrence was 30.0% over the next 10 years with no significant difference by etiology.

There was a nonsignificant increase in risk of recurrent SE between the two decades studied, which was present across all etiologies of SE. During the decade 1965 through 1974, the risk of recurrent SE was 26.2% compared with 31.8% during 1975 through 84 (log rank $p = 0.2$). The increase in recurrence risk was not explained by a differing distribution of etiology across the two decades ($p = 0.4$).

The risk of recurrence was 100% for progressive symptomatic SE, 23.6% for remote symptomatic SE, 26.1% for idiopathic/cryptogenic SE, and 26.3% for acute symptomatic SE (log rank $p = 0.09$). Age (log rank $p = 0.3$) and duration of SE (log rank $p = 0.3$) were not associated with the risk of recurrence within all status cases or within etiologic categories.

Among individuals with an unprovoked seizure before the initial episode of SE, the risk for recurrent SE was 31.2%. Among individuals with a first un-

Figure Cumulative risk for recurrent status epilepticus in Rochester, MN



Vertical lines show the pattern of censoring.

provoked seizure on the date of the initial SE, the risk for recurrent SE was 31.1%.

The risk of recurrence was significantly greater for females compared with males (39.3 vs 24.6%, log rank $p = 0.02$). This trend remained for individuals with acute symptomatic SE (28.3% for females vs 24.1% for males, log rank $p = 0.2$) and was significant for those with unprovoked SE (55.8% for females vs 25.7% for males, log rank $p = 0.048$).

Recurrent SE was less common among individuals with partial SE that did not generalize than among individuals with generalized tonic-clonic SE (20.5 vs 42.8%, log rank $p = 0.047$). This was due to the effect of seizure type on recurrence among individuals with acute symptomatic SE (14.6% for partial SE that did not generalize vs 39.3% for generalized tonic-clonic SE, log rank $p = 0.046$).

Therapeutic response to the first antiepileptic drug used to treat SE was associated with a reduced risk of recurrence (14.7% for those with a therapeutic response vs 38.2% for those without a therapeutic response, log rank $p = 0.01$). This trend remained but was no longer significant in the subgroups with acute symptomatic SE or unprovoked SE.

After examining the univariate effect of predictors of recurrence in separate Cox models, we created a final model based upon predictors that were significant in the univariate analyses (table). These variables included sex, etiology, seizure type, and therapeutic response to the first anticonvulsant drug. The risk of recurrent SE was increased 2.3-fold for females compared with males (95% CI = 1.1 to 5.0). Individuals with progressive symptomatic SE were 2.4-fold more likely than those with idiopathic/cryptogenic SE to recur, although this risk was no longer significant in the multivariate model (95% CI = 0.6-8.9). Compared to generalized tonic-clonic seizures without a partial compo-

Table Crude and adjusted rate ratios (RRs) and 95% CIs for association between status epilepticus (SE) and subsequent recurrent SE

	Crude		Adjusted*	
Variable	OR	95% CI	OR	95% CI
Therapeutic response to first AED				
Yes	0.4	0.2-0.9	0.3	0.1-0.7
No	1.0	Referent	1.0	Referent
Sex				
Female	2.3	1.1-4.7	2.3	1.1-5.0
Male	1.0	Referent	1.0	Referent
Etiology				
Remote symptomatic	1.3	0.4-4.2	1.5	0.4-5.3
Progressive symptomatic	3.5	1.1-11.2	2.4	0.6-8.9
Acute symptomatic	1.4	0.5-3.9	1.7	0.5-5.2
Idiopathic/cryptogenic	1.0	Referent	1.0	Referent
Seizure type				
Generalized with focal features	0.7	0.2-1.7	0.9	0.3-2.6
Partial only	0.43	0.19-0.97	0.5	0.2-1.1
Absence	0.7	0.2-3.2	0.8	0.1-4.5
Incomplete myoclonic	2.1	0.3-16	2.6	0.3-24
Generalized convulsive only	1.0	Referent	1.0	Referent
Duration, h				
≥24	1.7	0.7-4.0	Not included	
2-24	0.9	0.4-2.0		
<2	1.0	Referent		
Age, y				
<1	1.5	0.5-4.7	Not included	
1-20	1.0	Referent		
20-60	0.8	0.2-2.7		
>60	1.8	0.8-4.5		
Decade				
1975-84	1.5	0.7-3.3	Not included	
1965-74	1.0	Referent		

*Adjusted for all other variables in the model.

nent, partial seizures that did not generalize were protective for recurrence (rate ratio [RR] = 0.5, 95% CI = 0.21 to 1.1). Therapeutic response to the first antiepileptic drug for SE was protective for recurrence (RR = 0.3, 95% CI = 0.1 to 0.7). Several factors did not significantly alter the risk for recurrent SE, including duration, age, and decade of the first episode of SE.

DISCUSSION In this population-based study, about one-third of individuals experienced a second episode of SE with a doubling in risk for those with progressive symptomatic etiology and for females. The risk of recurrence was decreased for individuals with partial seizures that did not generalize and

when there was a positive therapeutic response to the first antiepileptic drug used to treat SE. There was no influence on recurrence risk whether the first unprovoked seizure occurred prior to the initial SE or was concurrent with the initial SE.

Others have reported the risk for recurrent SE over a 1- or 2-year period in children.⁹⁻¹¹ Among children, the recurrence risk at 1 year was 11%⁹ and 16%¹⁰ and was 18% at 2 years.⁹ Both of these studies included children with a first febrile SE, whereas we included only afebrile SE, and both report an increased risk for recurrence associated with pre-existing neurologic abnormality.^{9,10} In the one study of children followed for a mean of 6.3 years,¹² 20.8% of 24 children with an initial episode of SE

experienced recurrent SE. Among children in our study, the risk for recurrent afebrile SE was 13.1% at 1 year and 16.9% at 2 years, consistent with these prior studies. In the Richmond study,¹¹ which included both adults and children, the proportion with recurrent SE was 13.3% after 2 years of follow-up, consistent with the proportion of recurrence at 2 years in our study.

The occurrence of unprovoked seizure may modify the risk for recurrent SE. We examined this in our study, but failed to find a difference in recurrence according to the presence of a history of unprovoked seizure. Others have examined whether the risk of recurrent SE in children is modified by the presence of epilepsy.^{12,18} In one study of children with epilepsy, the risk for recurrent SE was 53% over a 10-year period, whereas in another study,¹⁸ there was a 32.1% risk of recurrent SE among children with SE prior to the diagnosis of epilepsy, although some children had multiple prior episodes of SE.

In our study, females were more likely to experience recurrent SE than males. This difference was not accounted for by differential mortality or loss to follow-up among males, nor was it accounted for by etiology of SE. This result requires further exploration in future studies.

The risk for recurrence was higher in the decade 1975 through 1984 than in the decade 1965 through 1974, although the difference was not significant. We examined whether this increased recurrence risk might have been due to differential ascertainment of recurrence by excluding those with nonconvulsive SE and those with SE in association with anoxic encephalopathy. We still found an increased risk for recurrence in the later decade. Detection bias is therefore an unlikely explanation for the time trend in recurrence. In our previous work,¹⁹ we found a decrease in case fatality during the decade from 1975 through 1984 after excluding individuals with anoxic encephalopathy. Thus, it is possible that improved survivorship after SE increased the number of people at risk for recurrent SE. If survivorship continues to improve, the risk for recurrence may also continue to increase.

We did not find an influence of unprovoked seizures occurring either before or concurrent with the initial episode of SE on risk for recurrence, nor did we find an influence of etiology, except for those with progressive symptomatic etiology. We were unable to measure time to initiation of treatment, but found that therapeutic response to the first antiepileptic drug used to treat SE was associated with a reduced risk of recurrence. Treatments available during our study period included barbiturates, phe-

nytoin, and diazepam, the same front-line treatments for SE that are used currently. These findings suggest that there is a subgroup of individuals with SE who are less responsive to drugs and are also predisposed to experience prolonged seizures as indexed by their increased risk for SE recurrence.

Approximately one-third of people with a first episode of afebrile SE will experience a recurrence over a 10-year period. Except for SE occurring in the setting of a progressive brain disorder, the risk of recurrence is about 25%, regardless of the underlying etiology or the age at first SE. This suggests that it is the tendency to experience SE itself and not the underlying etiology that is associated with recurrent SE episodes. Female gender and lack of therapeutic response to the first antiepileptic drug at the initial episode of SE identify those individuals at greatest risk for recurrence. Additionally, the proportion with recurrence increases with each additional episode of SE. Based on these data, therapies should be targeted to reduce the risk of SE recurrence among those at the highest risk.

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